Functions I

1. Find the domain and range of the following:

$$y = 5x^2 + 2,\ y = \sqrt{x^2 - 8x + 15},\ y = \tan x,\ y = \frac{1}{\cos x},\ y = \frac{2x+7}{4x-5}.$$

2. Let $f(x) = 2x^2 - 5x + 9$. Compute

$$f(2x),\ 3f(x),\ f(ix),\ 2f(\frac{1}{3}x - 5).$$

3. Let $f(x) = 2x^2 + 7x - 15$ and $g(x) = x^2 - 1$. Compute

$$f + g,\ f - g,\ f \circ g,\ f(g(1)), (g/f)(1).$$

4. Let $F(x) = 3x$. **(a)** Show that $F(x_1 + x_2) = F(x_1) + F(x_2)$. **(b)** Show that $F(cx_1) = cF(x_1)$. **(c)** Show that in general, if $L(x) = mx$, then $L(x_1 + x_2) = L(x_1) + L(x_2)$ and $L(cx_1) = cL(x_1)$.

5. Let $f(x) = x^2$. Graph

$$f(x),\ f(x) - 2,\ f(x+2),\ -2f(x+2),\ \frac{1}{2}f(x-2) + 2$$

6. Suppose some function f has a root at $x = 2$. Determine the roots of

$$-f(x),\ f(f(x)),\ f(x^2),\ 3f(x),\ f(x^3 - 2),\ f(\frac{x}{2}).$$

Write "UND" if it cannot be determined.

7. Find the average rates of change on the interval $[3, 12]$ for

$$f(x) = -3x + 2,\ f(x) = x^2 - 11x + 24,\ f(x) = \sqrt{x^3 - x^2 + 12}.$$

8. **(a)** A function f is said to be *even* if $f(-x) = f(x)$ for all x in its domain. Show that if (x, y) is a point on the graph of an even function f, then so is $(-x, y)$. **(b)** A function f is said to be *odd* if $f(-x) = -f(x)$ for all x in its domain. Show that if (x, y) is a point on the graph of an odd function f, then so is $(-x, -y)$. **(c)** Is each function (with domain $\mathbb{R}$) even, odd, or neither?

$$a(x) = x^3,\ b(x) = x^2 - 2,\ c(x) = x^3 + x^2,\ d(x) = \sin(x),\ e(x) = |x|$$

9. Suppose the position (height) of a falling object measured in feet follows the equation

$$h(t) = -16t^2 + 144$$

where t is measured in seconds. **(a)** When does the object hit the ground? **(b)** Find a general expression for the average rate of change $\Delta h/\Delta t$ over the interval $[1/2, 1/2{+}x]$. **(c)** What is a physical interpretation of the expression you found in part (b)? **(d)** Evaluate your expression as x tends to 0. What is a physical interpretation of the resulting expression?

10. Definition. For a function F and a fixed input x, define the *difference quotient*

$$A(F, h, x) = \frac{F(x+h) - F(x)}{h}, \quad h > 0.$$

(a) Let f and g be functions and c a real constant. Recall that $(f + g)(x) = f(x) + g(x)$ and $(cf)(x) = c \cdot f(x)$. Show that for any x and any $h > 0$:

$$A(f+g,\, h,\, x) = A(f, h, x) + A(g, h, x),$$
$$A(cf,\, h,\, x) = c \cdot A(f, h, x).$$

(b) Let $f(x) = x^2$. Compute $A(f, h, x)$ and simplify your answer as much as possible. What value does $A(f, h, x)$ approach as h gets closer and closer to 0?
Hint: after simplifying in part (i), think about what happens to each term when h is very small.